Week 2 - Assignment 5: Prompt Engineering Iteration Log

Real FL-01 Task: Frame a Content Refresh Prioritization project as a Machine Learning task for the FlyRank ML Internship.

Version 0 - Naive Prompt

Prompt:
Help me frame my machine learning project.

Representative Output:
The response briefly explained ML but was generic and unrelated to the project.

Technique: Baseline only.

Observed Improvement: Too generic.

Remaining Limitation: Assign a role.

Version 1 - Role Assignment

Prompt:
You are an experienced Machine Learning mentor helping students complete the FlyRank ML Internship. Help me frame my machine learning project.

Representative Output:
The answer became more professional and focused on ML concepts.

Technique: Role Assignment.

Observed Improvement: More relevant advice.

Remaining Limitation: Still lacked project context.

Version 2 - Context & Motivation

Prompt:
You are an ML mentor. My project is Content Refresh Prioritization using clicks, impressions, CTR and average position. Help me frame it as an ML task.

Representative Output:
The model identified the project as a ranking problem and suggested meaningful targets.

Technique: Context & Motivation.

Observed Improvement: Specific recommendations.

Remaining Limitation: Needed better organization.

Version 3 - Few-shot Examples

Prompt:
Example: Spam Detection -> Classification. Movie Recommendation -> Ranking. Now determine the ML task for Content Refresh Prioritization.

Representative Output:
The model followed the examples and selected Ranking with clearer reasoning.

Technique: Few-shot Examples.

Observed Improvement: Reasoning improved.

Remaining Limitation: Output still unstructured.

Version 4 - Output Structure

Prompt:
Present the answer in a table with Business Problem, ML Task, Target, Metric and Why ML.

Representative Output:
The response became organized and easy to review.

Technique: Output Structure.

Observed Improvement: Readability improved.

Remaining Limitation: Could still miss logical flow.

Version 5 - Step Decomposition

Prompt:
1. Identify the business problem. 2. Select the ML task. 3. Define the target. 4. Recommend metrics. 5. Explain why ML beats rules. 6. Summarize.

Representative Output:
The response covered every required section in logical order.

Technique: Step Decomposition.

Observed Improvement: More complete and consistent.

Remaining Limitation: Could be personalized further.

Cross-Model Comparison

Claude: Produced a concise, well-structured response that closely followed the requested format.

ChatGPT: Produced a more detailed explanation with additional examples, but occasionally included information beyond the requested structure.

Conclusion: Claude was stronger for concise submission-ready responses, while ChatGPT was better for learning and deeper explanations.

Final Reusable Prompt

You are an experienced Machine Learning mentor helping students complete the FlyRank ML Internship.

Project Context: The project analyzes search performance metrics such as clicks, impressions, CTR, and average position to prioritize webpages for content refresh.

Complete these steps: Identify the business problem. Determine the ML task type. Define a prediction target or proxy. Recommend suitable evaluation metrics. Explain why Machine Learning is more appropriate than a fixed rule. Present the answer in a professional table suitable for an internship submission. Keep the explanation concise, practical, and avoid generic AI language.